University of Algeirs 1

Computer science Department

Image Captioning Project

Phase 1: Dataset Exploration, Preprocessing and Multimodal Representation

Academic Project Report

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Chapter 1

Introduction

1.1 General Background

The rapid development of Artificial Intelligence (AI) has significantly improved the ability of computer systems to understand and process visual information. One important research area within this field is computer vision, which focuses on enabling machines to interpret and extract meaningful information from images and videos.

Among the different computer vision tasks, **image captioning** aims to automatically generate natural language descriptions of images. Unlike traditional image classification, which assigns one or more labels to an image, image captioning attempts to produce a meaningful sentence that describes the objects, their attributes, and the relationships or interactions between them.

For example, given an image containing a person riding a bicycle on a street, an image captioning system should be able to generate a sentence such as:

> "A person is riding a bicycle along a street."

This task requires combining two major capabilities: visual understanding and natural language processing. Consequently, image captioning systems rely on techniques that connect visual information with textual representations.

In recent years, multimodal models have become particularly important because they can represent information from different modalities, such as images and text, within a shared representation space.

1.2 Project Context

This project focuses on the development of an image captioning system based on a Retrieval-Augmented Generation (RAG) architecture.

The project uses the **Microsoft Common Objects in Context (MS COCO) 2017** dataset, which contains images of everyday scenes together with human-written captions and object annotations.

The main idea is to build a knowledge base containing images and their corresponding textual descriptions. This knowledge base can then be used by a retrieval mechanism to identify relevant information when processing a new input.

Our group, Group 29, is assigned the **Metadata-Filtered RAG** approach. This approach uses metadata and multimodal representations to support the retrieval process.

Before developing and evaluating the retrieval system, it is essential to prepare a reliable dataset. The quality, organization, and consistency of the dataset can directly affect the quality of the retrieval results and the overall system performance.

Therefore, this report focuses on the first phase of the project: dataset exploration, selection, preprocessing, quality verification, and multimodal representation.

1.3 Problem Statement

Building an effective image captioning system requires more than simply collecting images and their captions. The dataset must be carefully examined and prepared to ensure that it is suitable for the intended task.

Several challenges may arise during this preparation process, including:

- **Dataset organization:** The original dataset contains a large number of images, annotations, and captions that must be organized into a consistent structure.
- **Data consistency:** Images may have different numbers of associated captions. A consistent caption selection policy is therefore necessary.
- **Scene diversity:** The selected images should represent different levels of scene complexity, object categories, and visual interactions.
- **Data quality:** The selected images and their annotations must be checked to identify inconsistencies or unsuitable samples.
- **Text preprocessing:** Captions must be standardized before being converted into numerical representations for retrieval.
- **Multimodal representation:** Images and captions must be encoded into compatible vector representations to support similarity-based retrieval.
- **Data leakage prevention:** Training, validation, and testing data must be separated appropriately to ensure a reliable evaluation.

The central problem addressed in this phase is therefore:

How can the original MS COCO 2017 dataset be explored, selected, cleaned, and transformed into a consistent multimodal knowledge base suitable for an image captioning system based on Metadata-Filtered RAG?

1.4 Objectives of Phase 1

The main objective of this phase is to prepare a structured and reliable dataset that can be used in the subsequent development of the image captioning system.

The specific objectives are:

1. Examine the organization and characteristics of the MS COCO 2017 dataset.
2. Define measurable criteria for identifying different levels of scene complexity.
3. Construct a multi-label representation of image categories to support stratified sampling.
4. Select a diverse subset of 100 images from the official validation dataset for testing.
5. Apply a consistent caption selection strategy to obtain exactly five captions per training image.
6. Perform visual quality assurance on the selected test images.
7. Preprocess the textual captions while preserving their original versions.
8. Prepare a validation subset and perform exploratory data analysis.
9. Generate image and text embeddings using a CLIP-based model.
10. Save the resulting data and embeddings in suitable formats for subsequent retrieval operations.

1.5 Dataset Overview

The project uses the MS COCO 2017 dataset, which provides images, object annotations, and natural language captions.

Two main subsets are considered during this phase:

1.5.1 COCO Train2017

The training subset is used to construct the knowledge base of the RAG system.

According to the project specifications, the knowledge base is designed to contain:

- 10,000 selected training images.
- 50,000 corresponding captions.
- Five captions associated with each image.

These images and captions provide the visual and textual information required for the retrieval process.

1.5.2 COCO Val2017

The official validation subset contains 5,000 images. It is used in this phase to construct a test subset consisting of 100 images.

The selected images are intended to cover different visual situations, including:

- Simple scenes.
- Complex scenes.
- Images containing multiple objects.
- Interactions between objects or people.
- Different visual contexts.

The selected test subset is intended to be identical across all project groups, as specified by the professor.

1.6 Scope of the Work

This report documents the dataset preparation phase rather than the complete implementation of the image captioning system.

The work covered in this phase includes:

- Environment and storage preparation.
- Dataset inspection and organization.
- Scene complexity analysis.
- Multi-label matrix construction.

- Stratified test image selection.
- Caption selection and standardization.
- Visual quality assurance.
- Text preprocessing.
- Exploratory data analysis.
- Image and text embedding generation.

The subsequent phases will use these prepared resources to develop and evaluate the Metadata-Filtered RAG system.

1.7 Report Organization

This report is organized into several chapters:

- **Chapter 1 – Introduction:** Presents the project background, context, problem statement, objectives, dataset overview, and scope.
- **Chapter 2 – Dataset Description and Project Specifications:** Describes the MS COCO dataset and the requirements that guide the preparation process.
- **Chapter 3 – Methodology and Implementation:** Explains the operations performed during Phase 1, including dataset inspection, sampling, caption processing, and embedding generation.
- **Chapter 4 – Results and Quality Assurance:** Presents the obtained results, visual inspections, statistical observations, and verification procedures.
- **Chapter 5 – Conclusion:** Summarizes the work completed during this phase and presents the prepared resources for the next stage of the project.

Chapter 2

Dataset Description and Project Specifications

2.1 Introduction

The quality of an image captioning system depends heavily on the quality and organization of the data used during its development. Therefore, understanding the structure, content, and characteristics of the selected dataset is an essential preliminary step.

This chapter presents the MS COCO 2017 dataset used in the project, describes its main components, and explains the specifications established for preparing the data.

Particular attention is given to image annotations, natural language captions, scene complexity, and the organization of the training, validation, and testing subsets.

2.2 MS COCO 2017 Dataset

2.2.1 General Overview

Microsoft Common Objects in Context (MS COCO) is a large-scale dataset designed to support research in computer vision and visual understanding.

It contains everyday scenes captured in different contexts, with multiple objects appearing in the same image. The dataset provides detailed object annotations and human-written descriptions, making it suitable for tasks such as object detection, image segmentation, and image captioning.

The 2017 version of MS COCO is used in this project because it provides the visual and textual information required to construct a multimodal knowledge base.

2.2.2 Main Dataset Components

The dataset provides several types of information. The components particularly relevant to this project are described below.

- **Images:** The visual data consists of photographs representing everyday scenes, objects, people, and interactions.
- **Object annotations:** These annotations describe the objects present in images. They include category information and bounding-box coordinates, which can be used to analyze the visual content of each image.
- **Image captions:** Captions are natural language descriptions written by human annotators. They provide semantic information about the content of each image.
- **Category information:** COCO defines 80 object categories used to identify and classify objects appearing in the images.

The combination of these components allows each image to be represented through both visual information and associated textual and categorical metadata.

2.3 Dataset Subsets Used in the Project

Two official subsets of MS COCO 2017 are used during the dataset preparation phase: Train2017 and Val2017.

2.3.1 COCO Train2017

The Train2017 subset is the source of the images and captions used to construct the knowledge base.

According to the project specifications, the knowledge base must contain:

- 10,000 selected images.
- Five captions for each selected image.
- A total of 50,000 captions.

The selected images and their associated captions constitute the main visual and textual resources available to the retrieval system.

In addition, a separate validation subset of 100 images must be prepared from Train2017. These images are intended for validation and hyperparameter tuning.

The validation images must be different from the images included in the knowledge base to avoid overlap between the training and validation data.

2.3.2 COCO Val2017

The official Val2017 subset contains 5,000 images. It serves as the source for selecting the test subset used to evaluate the system.

The project requires the selection of 100 images using a stratified sampling strategy.

The selected test subset must be identical for all project groups to ensure that their systems are evaluated under the same conditions.

The test subset must also contain 500 reference captions, corresponding to five ground-truth captions per image.

2.3.3 Summary of Dataset Organization

The main dataset requirements are summarized in Table 2.1.

Table 2.1: Dataset organization and project requirements

Subset	**Source**	**Number of Images**	**Purpose**
Knowledge Base	Train2017	10,000	Training and retrieval knowledge base
Validation	Train2017	100	Hyperparameter tuning and validation
Test	Val2017	100	Final evaluation of the system

For the knowledge base, the expected number of captions is 50,000, while the test subset contains 500 reference captions.

2.4 Image Annotations and Object Categories

2.4.1 Object Annotations

In addition to the visual content, COCO provides structured annotations describing the objects present in each image.

Each object annotation contains information such as:

- The identifier of the associated image.
- The category identifier of the object.
- The bounding-box coordinates.
- Additional annotation information.

Bounding boxes describe the approximate location of objects within an image and are represented by four values:

$$B = (x, y, w, h) \tag{2.1}$$

where:

- x and y represent the coordinates of the upper-left corner.
- w represents the width of the bounding box.
- h represents its height.

These annotations provide structured information that can be used to analyze the number and diversity of objects appearing in each scene.

2.4.2 The 80 Object Categories

MS COCO defines 80 object categories, including people, vehicles, animals, furniture, and everyday objects.

Examples include:

- Person
- Bicycle
- Car
- Motorcycle
- Bus
- Bottle
- Chair
- Dining table
- Dog
- Laptop

Each category is associated with an identifier. These identifiers are not necessarily consecutive, so a mapping between category identifiers and matrix columns is required when constructing a numerical representation of the categories.

2.5 Image Scene Complexity

2.5.1 Importance of Scene Complexity

Images do not all contain the same level of visual complexity.

Some images contain only a few objects and simple visual arrangements, while others contain many objects, multiple categories, and interactions between different elements.

For this project, scene complexity is an important consideration when selecting the test images.

A test subset containing only simple scenes would not provide a sufficiently representative evaluation of the system's ability to understand more complicated visual situations.

Therefore, the project requires images covering different levels of complexity.

2.5.2 Criteria Used for Scene Classification

To make scene complexity measurable, two features can be extracted from the object annotations:

1. **Number of objects:** The total number of annotated object instances in an image.
2. **Number of unique categories:** The number of distinct object categories represented in that image.

Let:

$$O_i = \text{Number of objects in image } i \tag{2.2}$$

$$C_i = \text{Number of unique categories in image } i \tag{2.3}$$

These two measurements provide a simple way to characterize the visual complexity of each scene.

For the implementation in this project, the following criteria were used:

- **Simple scenes:** Images containing no more than three annotated objects and no more than two unique categories.
- **Complex scenes:** Images containing at least eight annotated objects or at least four unique categories.
- **Moderate scenes:** Images that do not satisfy the simple or complex conditions.

These criteria transform the qualitative concept of scene complexity into measurable characteristics that can support the image selection process.

2.6 Multi-Label Representation

2.6.1 Concept

Each image may contain objects belonging to several different categories.

For example, an image may contain a person, a bicycle, and a car simultaneously. Therefore, representing an image using only one category would result in the loss of important information.

To address this issue, a multi-label representation is used.

In this representation, each image is associated with a binary vector containing 80 elements, where each element corresponds to one COCO object category.

2.6.2 Multi-Label Matrix

The complete set of image representations can be organized into a binary matrix:

$$Y \in \{0,1\}^{N\times 80} \tag{2.4}$$

where:

- N represents the number of images.
- 80 represents the number of COCO categories.
- $Y_{ij} = 1$ indicates that category j appears in image i.
- $Y_{ij} = 0$ indicates that category j does not appear in image i.

This representation allows the distribution of object categories across images to be analyzed and supports multi-label stratified sampling.

2.7 Caption Data and Standardization

2.7.1 Human-Written Captions

The COCO caption annotations provide natural language descriptions of the visual content of images.

Unlike object category labels, which provide short categorical information, captions describe scenes using complete sentences.

An image can therefore have several different descriptions, each emphasizing different aspects of its visual content.

These descriptions are particularly important for image captioning and text-based retrieval.

2.7.2 Five-Caption Requirement

The project specifications require exactly five captions for each image included in the knowledge base.

This requirement ensures consistency in the number of textual descriptions associated with each image.

For 10,000 knowledge-base images, the expected number of captions is:

$$10,000 \times 5 = 50,000 \tag{2.5}$$

Consequently, a caption selection policy is necessary when the original annotations contain more than five captions for certain images.

2.7.3 Caption Preservation

The original captions should be preserved to maintain the integrity of the source annotations.

If text preprocessing is applied, the processed captions should be stored separately from the original versions.

This separation allows the system to use standardized text for retrieval while retaining the original human-written descriptions for reference and traceability.

2.8 Multimodal Representation and CLIP

2.8.1 From Raw Data to Numerical Representations

Images and textual captions are fundamentally different types of data.

Images consist of visual information represented by pixel values, whereas captions consist of sequences of words.

To compare these two modalities using similarity-based retrieval, they must be transformed into numerical representations called embeddings.

An embedding is a vector of numbers that encodes meaningful information about an input.

2.8.2 Contrastive Language–Image Pre-training

CLIP (Contrastive Language–Image Pre-training) is a multimodal model designed to learn relationships between images and natural language descriptions.

It uses image and text encoders to transform both modalities into vectors within a shared embedding space.

As a result, semantically related images and captions can be represented by vectors that are close to each other in this space.

This property makes CLIP suitable for applications involving image-text similarity and multimodal retrieval.

2.8.3 Embedding Similarity

Once image and text embeddings have been generated, their similarity can be measured using cosine similarity.

For two vectors A and B, cosine similarity is defined as:

$$\text{sim}(A, B) = \frac{A \cdot B}{\|A\|\|B\|} \tag{2.6}$$

where:

- $A \cdot B$ represents the dot product of the two vectors.
- $\|A\|$ and $\|B\|$ represent their Euclidean norms.

A higher cosine similarity indicates a greater degree of alignment between the two representations.

In this project, image and text embeddings are prepared to support the retrieval mechanism of the Metadata-Filtered RAG architecture.

2.9 Metadata-Filtered RAG Architecture

2.9.1 General Concept

Retrieval-Augmented Generation (RAG) is an approach that combines information retrieval with a generation model.

Instead of relying exclusively on information learned during model training, a RAG system retrieves relevant information from an external knowledge base and uses it as additional context.

The Metadata-Filtered RAG approach assigned to Group 29 incorporates metadata into the retrieval process.

Metadata can provide additional information about the images, such as object categories and other descriptive characteristics.

2.9.2 Role of Metadata and Embeddings

The knowledge base prepared during Phase 1 provides two complementary types of information:

- **Embeddings:** Numerical representations used to measure semantic similarity between images and text.
- **Metadata:** Structured information that can be used to filter or refine the retrieval results.

Combining these elements can help the retrieval system identify relevant information more efficiently.

The preparation of reliable metadata and compatible multimodal embeddings is therefore an important foundation for the subsequent implementation phase.

2.10 Data Separation and Leakage Prevention

A reliable evaluation requires a clear separation between the knowledge base, validation data, and test data.

Data leakage occurs when information from the evaluation data becomes accessible during training or retrieval in a way that could artificially improve the evaluation results.

To avoid this problem, the project establishes the following requirements:

- The 100 validation images must be different from the 10,000 knowledge-base training images.
- The 100 test images must come from Val2017.
- Test reference captions must not be included in the knowledge base.
- The same test subset must be used by all project groups.

Maintaining these separations is essential for ensuring that the evaluation measures the system's ability to process previously unseen images rather than retrieve information that has already been included in its knowledge base.

2.11 Chapter Summary

This chapter introduced the MS COCO 2017 dataset and the main requirements governing its use in the project.

It presented the dataset's images, object annotations, 80 object categories, and human-written captions. It also explained the importance of scene complexity, multi-label representation, caption standardization, and multimodal embeddings.

Finally, the chapter described the principles of Metadata-Filtered RAG and the importance of separating training, validation, and testing data.

These concepts establish the theoretical and technical foundation for the dataset preparation methodology presented in the following chapter.

Chapter 3

Methodology and Implementation

3.1 Introduction

This chapter describes the practical methodology followed during Phase 1 of the project. It presents the different operations performed to transform the original MS COCO 2017 dataset into structured resources suitable for the Metadata-Filtered RAG system.

The implementation was carried out using Python in Google Colab, with Google Drive used for persistent storage of project files and generated results.

The methodology consists of eight main steps:

1. Environment recovery and storage verification.
2. Scene complexity analysis and multi-label matrix construction.
3. Stratified selection of the test subset.
4. Standardization of the five-caption rule.
5. Visual quality assurance.
6. Text preprocessing.
7. Validation subset preparation and exploratory data analysis.
8. Multimodal embedding generation.

Each step is described according to its objective, methodology, and obtained results.

3.2 Step 1: Environment Recovery and Storage Verification

3.2.1 Objective

The first step consisted of recovering the project environment and verifying the availability of the previously downloaded dataset files.

Google Colab provides temporary runtime storage. Consequently, files saved in the temporary environment may disappear when the runtime is disconnected or reset.

To avoid repeatedly downloading large annotation files, Google Drive was used as persistent storage.

3.2.2 Methodology

The Google Drive environment was mounted in Colab, and the project directory was inspected to verify the presence of the required files.

The main working directory was:

```
/content/drive/MyDrive/COCO_Project/
```

The project directory was organized into the following subdirectories:

- `raw/`: Original downloaded data.
- `images/train/`: Training images.
- `images/test/`: Selected test images.
- `annotations/`: COCO annotation files.
- `cleaned/`: Processed dataset files.
- `reports/`: Reports and analysis outputs.
- `caption_selection/`: Caption selection results.
- `embeddings/`: Generated image and text embeddings.

The directory inspection also confirmed the persistence of the MS COCO Val2017 annotation files from previous sessions.

3.2.3 Result

The project environment was successfully recovered, and the required annotation resources were available in persistent storage.

This organization provided a structured working environment and reduced the need for repeated downloads after runtime interruptions.

Insert screenshot of the Google Drive project directory here.

Figure 3.1: Project directory structure in Google Drive

3.3 Step 2: Scene Complexity Analysis and Multi-Label Matrix

3.3.1 Objective

The second step aimed to analyze the visual characteristics of the 5,000 images in the official Val2017 subset.

The purpose was to identify images representing different levels of scene complexity and prepare a numerical representation of object categories for stratified sampling.

Since the professor required the test subset to cover simple scenes, complex scenes, multiple objects, and different visual contexts, measurable criteria were needed to support the selection process.

3.3.2 Scene Complexity Analysis

Two features were extracted from the object annotations of each image:

- The total number of annotated object instances.
- The number of unique object categories.

Let O_i represent the number of objects in image i, and C_i represent the number of unique categories.

The following classification criteria were defined:

Simple scenes:

$$O_i \leq 3 \quad \text{and} \quad C_i \leq 2 \tag{3.1}$$

Complex scenes:

$$O_i \geq 8 \quad \text{or} \quad C_i \geq 4 \tag{3.2}$$

Moderate scenes:

Images that satisfy neither the simple nor the complex conditions.

These rules transform the qualitative concept of scene complexity into measurable characteristics.

3.3.3 Construction of the Multi-Label Matrix

In addition to scene complexity, the distribution of object categories was considered.

Each image may contain multiple categories. Therefore, a multi-label representation was constructed to describe the presence or absence of each of the 80 COCO categories.

The resulting matrix is represented as:

$$Y \in \{0,1\}^{5000\times 80} \tag{3.3}$$

Each row corresponds to one image, while each column corresponds to one object category.

The matrix elements are defined as:

$$Y_{ij} = \begin{cases} 1 & \text{if category } j \text{ appears in image } i, \\ 0 & \text{otherwise.} \end{cases} \tag{3.4}$$

This representation allows the category distribution to be analyzed across the complete Val2017 subset.

3.3.4 Result

The analysis produced a binary multi-label matrix with 5,000 rows and 80 columns.

This matrix was subsequently used as the input for the iterative stratification procedure.

The use of a multi-label representation made it possible to consider the presence of several object categories within the same image during the selection process.

3.4 Step 3: Stratified Selection of the Test Subset

3.4.1 Objective

The objective of this step was to select 100 images from the 5,000 images in Val2017.

The selected subset needed to represent different object categories and levels of scene complexity, rather than containing only images from a limited number of visual situations.

3.4.2 Iterative Multi-Label Stratification

To support the selection process, the `MultilabelStratifiedShuffleSplit` method from the iterative-stratification library was used.

Unlike ordinary random sampling, multi-label stratification considers the distribution of several labels simultaneously.

In this project, the 80 object categories were represented by the columns of the previously constructed multi-label matrix.

The stratification process aimed to select images while approximately preserving the distribution of the object categories.

This approach is particularly relevant because an image can contain several categories at once.

3.4.3 Final Selection

During the selection procedure, an initial candidate set of 106 images was obtained.

Six images were subsequently removed using a fixed random seed of 42 to obtain the required final subset of 100 images.

The resulting scene-complexity distribution was:

- Simple scenes: 40 images.
- Moderate scenes: 18 images.
- Complex scenes: 42 images.

The final distribution is summarized in Table 3.1.

Table 3.1: Scene complexity distribution in the selected 100-image test subset

Scene Category	**Number of Images**	**Percentage**
Simple	40	40%
Moderate	18	18%
Complex	42	42%
Total	**100**	**100%**

3.4.4 Result

The final subset contains images from three levels of scene complexity.

The selected images were intended to provide visual variety and support the evaluation of the system under different scene conditions.

The selected image identifiers should be preserved to ensure reproducibility and to facilitate the use of the same test subset by all project groups.

Insert screenshot of the test subset scene distribution here.

Figure 3.2: Scene complexity distribution of the selected test images

3.5 Step 4: Standardization of the Five-Caption Rule

3.5.1 Objective

The project specifications require exactly five captions for each image included in the training knowledge base.

However, the original COCO caption annotations do not contain exactly the same number of captions for every image.

Therefore, a consistent selection strategy was required to obtain a standardized dataset.

3.5.2 Initial Caption Analysis

The original Train2017 caption annotations were examined to determine the number of captions associated with each image.

The analysis produced the following results:

- Total number of images: 118,287.
- Total number of caption annotations: 591,753.
- Images with five captions: 117,972.
- Images with six captions: 312.
- Images with seven captions: 3.

The distribution is presented in Table 3.2.

Table 3.2: Original caption distribution in Train2017

Number of Captions per Image	**Number of Images**
5	117,972
6	312
7	3
Total	**118,287**

3.5.3 Caption Selection Strategy

A deterministic selection strategy, referred to as Strategy A, was adopted.

For each image, its associated captions were sorted according to their official annotation identifiers.

When an image contained more than five captions, only the first five captions in this ordering were retained.

This strategy was selected because it is:

- Deterministic and reproducible.
- Easy to implement and verify.
- Independent of subjective human quality judgments.
- Consistent with the requirement of five captions per image.

It should be noted that sorting by annotation identifier does not imply that the retained captions are objectively better than the excluded captions. The strategy provides a consistent selection rule rather than a quality ranking.

3.5.4 Final Caption Count

The number of excluded annotations was calculated as:

$$E = (312 \times 1) + (3 \times 2) \tag{3.5}$$

$$E = 318 \tag{3.6}$$

Therefore, the final number of captions becomes:

$$591753 - 318 = 591435 \tag{3.7}$$

This result corresponds exactly to:

$$118287 \times 5 = 591435 \tag{3.8}$$

Thus, the processed training caption collection contains exactly five captions per image.

For the selected 10,000-image knowledge base, the corresponding target is 50,000 captions.

3.5.5 Result

The caption selection strategy successfully established a consistent five-caption structure across the analyzed Train2017 annotations.

This standardization simplifies dataset organization and ensures consistency when constructing the knowledge base.

3.6 Step 5: Visual Quality Assurance

3.6.1 Objective

Numerical statistics and category distributions alone cannot fully describe the visual characteristics of a dataset.

For this reason, a visual inspection was performed to examine the selected test images and verify their diversity.

3.6.2 Contact Sheet Generation

The 100 selected Val2017 images were downloaded from the official MS COCO image repository.

A contact sheet was then generated by arranging the images in a grid containing 10 rows and 10 columns.

This representation allows all selected images to be inspected simultaneously.

The visual inspection aimed to identify:

- Differences in scene complexity.
- Variety in object categories.
- Scenes containing multiple objects.
- Different visual contexts.
- Interactions between people and objects.

3.6.3 Result

A 10×10 contact sheet containing all 100 selected images was generated and saved as a JPEG file.

This visual representation provided an additional quality assurance method alongside the numerical analysis performed during the selection process.

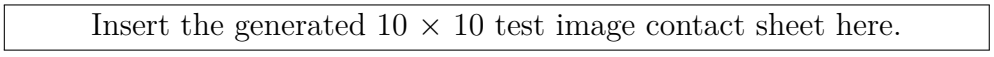


Figure 3.3: Contact sheet of the 100 selected Val2017 images

3.7 Step 6: Text Preprocessing for Retrieval

3.7.1 Objective

Before generating text embeddings, the captions were normalized to reduce superficial differences in their textual representation.

The objective was to prepare standardized textual data for the retrieval process while preserving the original human-written captions.

3.7.2 Preprocessing Operations

The preprocessing procedure included three main operations:

1. **Lowercasing:** All alphabetic characters were converted to lowercase.
2. **Punctuation removal:** Punctuation characters were removed to reduce superficial textual variations.
3. **Whitespace normalization:** Multiple spaces and unnecessary whitespace characters were replaced with standardized spacing.

For example, the following original caption:

"A Man Riding A Bicycle, Along The Street!"

becomes:

"a man riding a bicycle along the street"

3.7.3 Preservation of Original Captions

The original captions were not overwritten.

Instead, two separate fields were maintained:

- `raw_captions`: Original human-written descriptions.
- `clean_captions`: Normalized versions prepared for retrieval.

This separation preserves the original annotations while allowing the retrieval system to use standardized textual representations.

3.7.4 Result

The 50,000 training captions were processed using the defined normalization operations.

The cleaned captions were stored separately from their original versions, maintaining the relationship between the source annotations and the processed text.

3.8 Step 7: Validation Subset Preparation and Exploratory Data Analysis

3.8.1 Objective

The objective of this step was to prepare a validation subset and examine basic statistical characteristics of the selected training data.

Exploratory Data Analysis (EDA) was performed to obtain a better understanding of the dataset before the subsequent retrieval implementation.

3.8.2 Validation Subset Selection

A subset of 100 images was randomly sampled from the previously selected 10,000 Train2017 image IDs.

A fixed random seed of 42 was used to make the sampling process reproducible.

The resulting subset was intended for validation and hyperparameter tuning.

Important separation requirement: The project specifications require validation images to be different from the images included in the knowledge base.

Therefore, the 100 sampled image IDs must be excluded from the final knowledge-base image set. This separation must be verified in the implementation before the dataset can be considered compliant with the project specifications.

3.8.3 Exploratory Data Analysis

Histograms were generated to examine two characteristics of the dataset:

- Caption length, measured in number of words.
- Number of annotated objects per image.

The analysis produced the following reported averages:

- Average caption length: 10.46 words.
- Average number of objects per image: 7.26.

These measurements provide an overview of the textual and visual characteristics of the analyzed data.

3.8.4 Result

The validation subset was sampled using a fixed random seed, and statistical visualizations were generated to examine caption length and object count.

The histograms provide a useful overview of the distribution of these characteristics and can help identify potential irregularities in the dataset.

Insert caption length histogram here.

Figure 3.4: Distribution of caption lengths

Insert object count histogram here.

Figure 3.5: Distribution of annotated objects per image

3.9 Step 8: Multimodal Embedding Generation

3.9.1 Objective

The final step consisted of transforming images and their associated captions into numerical vector representations called embeddings.

These embeddings are essential for the similarity-based retrieval mechanism used in the Metadata-Filtered RAG architecture.

3.9.2 CLIP ViT-B/32 Model

A CLIP ViT-B/32 model was used to generate both image and text embeddings.

CLIP is a multimodal model that maps visual and textual inputs into a shared representation space.

This property allows the system to compare images and text through vector similarity.

The selected model produces embeddings with 512 dimensions.

3.9.3 Text Embedding Generation

The 50,000 training captions were encoded into text embeddings.

Each caption was transformed into a numerical vector containing 512 values.

Therefore, the expected text embedding matrix has the following dimensions:

$$E_{\text{text}} \in R^{50000 \times 512} \tag{3.9}$$

Each row represents one caption embedding.

3.9.4 Image Embedding Generation

The 10,000 selected training images were also encoded using the image encoder.

Each image was transformed into a 512-dimensional vector.

The expected image embedding matrix has the following dimensions:

$$E_{\text{image}} \in R^{10000 \times 512} \tag{3.10}$$

Each row corresponds to one image embedding.

3.9.5 Batch Processing

Image embedding generation was performed in batches of 500 images.

Batch processing was adopted to reduce memory consumption and avoid exceeding the available Google Colab runtime resources.

Instead of processing all 10,000 images simultaneously, the images were divided into smaller groups.

The generated embeddings were then collected and saved for subsequent use.

3.9.6 Storage of Embeddings

The resulting image and text embedding arrays were saved in NumPy `.npy` format.

This format allows numerical arrays to be stored and loaded efficiently using Python.

The expected dimensions of the generated embeddings are summarized in Table 3.3.

Table 3.3: Expected dimensions of multimodal embeddings

Embedding Type	**Number of Vectors**	**Dimensions**
Text embeddings	50,000	512
Image embeddings	10,000	512

3.9.7 Result

The image and text embedding generation process produced numerical representations intended for the subsequent retrieval stage.

The vectors were saved in NumPy format and organized in the project's embedding directory.

These representations provide the multimodal foundation required to compare textual queries with visual content in the knowledge base.

3.10 Chapter Summary

This chapter presented the eight main steps performed during Phase 1 of the project.

The work began with the recovery of the Google Colab environment and verification of persistent storage.

The Val2017 dataset was then analyzed to measure scene complexity and construct a multi-label matrix representing the 80 COCO object categories.

A stratified sampling procedure was used to select 100 test images, followed by caption standardization, visual quality assurance, and text preprocessing.

A validation subset was also sampled, and exploratory data analysis was performed to examine caption length and object count.

Finally, image and text embeddings were generated using a CLIP ViT-B/32 model and saved in NumPy format.

The resulting resources form the basis for the next phase, which focuses on implementing and evaluating the Metadata-Filtered RAG retrieval system.

Chapter 4

Results and Quality Assurance

4.1 Introduction

After completing the dataset preparation operations described in the previous chapter, several results were obtained concerning the organization, consistency, and multimodal representation of the MS COCO dataset.

This chapter summarizes the main results of Phase 1 and presents the verification procedures performed to assess the quality of the prepared resources.

The evaluation focuses on dataset consistency, image diversity, caption standardization, text preprocessing, and embedding generation.

4.2 Summary of Dataset Preparation Results

The preparation process produced several resources required for the subsequent implementation of the Metadata-Filtered RAG system.

The main results are summarized in Table 4.1.

Table 4.1: Summary of Phase 1 results

Operation	**Obtained Result**
Val2017 analysis	5,000 images analyzed
Multi-label representation	Binary matrix of size 5000×80
Test subset selection	100 selected images
Test scene distribution	40 Simple, 18 Moderate, 42 Complex
Train2017 caption analysis	118,287 images and 591,753 annotations
Caption standardization	Exactly five captions per image
Standardized caption collection	591,435 captions across 118,287 images
Knowledge base target	10,000 images and 50,000 captions
Text preprocessing	50,000 captions processed
Image embeddings	10,000 image vectors, 512 dimensions each
Text embeddings	50,000 text vectors, 512 dimensions each

4.3 Test Subset Quality Assessment

4.3.1 Scene Complexity Distribution

The selected test subset contains 100 images distributed across three scene complexity categories.

The distribution is presented below:

Table 4.2: Distribution of test images by scene complexity

Category	**Number of Images**	**Percentage**
Simple	40	40%
Moderate	18	18%
Complex	42	42%
Total	100	100%

The selected subset includes both relatively simple scenes and scenes containing several objects or multiple object categories.

This distribution provides a range of visual situations for evaluating the image captioning system.

4.3.2 Visual Diversity Verification

The contact sheet generated during the visual quality assurance step allowed the selected images to be examined simultaneously.

The inspection complements the numerical analysis because category counts alone cannot fully describe the relationships and interactions represented in an image.

The visual inspection is intended to verify that the selected subset contains a variety of scenes and is not dominated by a single type of visual content.

4.4 Caption Consistency and Standardization

4.4.1 Initial and Final Caption Counts

The original Train2017 caption annotations contained 591,753 captions associated with 118,287 images.

After applying the five-caption selection strategy, 318 excess annotations were excluded.

The resulting caption count was:

$$591753 - 318 = 591435 \tag{4.1}$$

This corresponds to exactly five captions per image:

$$118287 \times 5 = 591435 \tag{4.2}$$

Therefore, the standardization procedure achieved the required five-caption structure across the analyzed training annotation collection.

4.4.2 Caption Consistency

The use of a deterministic selection policy ensures that the same five captions are retained whenever the procedure is repeated on the same source annotations.

Furthermore, preserving the original captions alongside their normalized versions allows the processed text to be traced back to its source.

This separation is useful for debugging, verification, and subsequent retrieval operations.

4.5 Exploratory Data Analysis Results

Exploratory Data Analysis was performed to examine the textual and visual characteristics of the prepared dataset.

Two main measurements were considered:

- Caption length, measured in number of words.

- Number of annotated objects per image.

The reported average values were:

- Average caption length: 10.46 words.
- Average number of objects per image: 7.26.

These measurements provide an initial description of the dataset.

The caption length distribution helps characterize the textual descriptions, while the object count distribution provides information about the density of annotated objects in the analyzed images.

The corresponding histograms are useful for observing the distribution and identifying unusually short or long captions and images with particularly high or low object counts.

Note: The population used to calculate these averages must be specified according to the actual notebook results (for example, the 10,000 selected training images or the 100 validation images).

4.6 Multimodal Embedding Results

4.6.1 Embedding Dimensions

The embedding generation process was designed to produce a numerical representation for each selected image and training caption.

The expected dimensions are summarized in Table 4.3.

Table 4.3: Multimodal embedding dimensions

Embedding Type	**Number of Vectors**	**Vector Dimension**
Image embeddings	10,000	512
Text embeddings	50,000	512

Each image or caption is represented by a 512-dimensional vector.

The use of a shared embedding space allows image and text representations to be compared through similarity measurements.

4.6.2 Storage and Organization

The generated embedding arrays were saved in NumPy `.npy` format.

This format allows the vectors to be stored and loaded without repeating the encoding process.

Image embeddings were generated using batches of 500 images to reduce memory consumption during processing.

The resulting files are intended to be reused during the implementation of the retrieval system.

4.7 Quality Assurance and Verification

Several verification procedures were considered during the preparation process.

Table 4.4 summarizes the main quality assurance criteria.

Table 4.4: Dataset quality assurance checklist

Verification Criterion	**Expected Condition**
Image identifiers	Unique and correctly associated with annotations.
Object categories	Consistent mapping to the 80 COCO categories.
Multi-label matrix	Correct dimensions and binary values.
Test subset	Exactly 100 selected images.
Scene diversity	Representation of simple, moderate, and complex scenes.
Caption consistency	Exactly five captions per knowledge-base image.
Text preprocessing	Cleaned captions available while original captions are preserved.
Embedding dimensions	512-dimensional image and text vectors.
Data separation	No overlap between knowledge-base, validation, and test images.

The final criterion is particularly important. The validation images must be excluded from the knowledge base, and test reference captions must not be used as retrieval knowledge for their corresponding test images.

These conditions should be verified by comparing the image identifiers and caption associations across the different subsets.

4.8 Technical Challenges and Solutions

During the preparation process, several technical challenges were encountered.

4.8.1 Temporary Runtime Storage

Google Colab uses temporary runtime storage, which can disappear after a session reset.

Solution: Google Drive was used to preserve the project files and avoid unnecessary repeated downloads.

4.8.2 Large Dataset Processing

Processing a large number of images simultaneously can exceed the available memory resources.

Solution: Image embedding generation was divided into batches of 500 images to reduce memory consumption.

4.8.3 Caption Count Inconsistency

Some Train2017 images contained more than five caption annotations.

Solution: A deterministic selection strategy was applied to retain exactly five captions per image.

4.8.4 Multi-Label Image Representation

Each image may contain several object categories, making ordinary single-label sampling unsuitable for representing the complete category distribution.

Solution: A binary multi-label matrix was constructed and used with an iterative stratification method.

4.9 Limitations and Considerations

Although the dataset preparation process produced the resources required for the next phase, several limitations should be considered.

- The scene complexity criteria are based on object counts and category diversity. They do not fully capture semantic relationships or the difficulty of understanding an image.
- Selecting the first five captions according to annotation identifiers ensures reproducibility, but does not guarantee that the selected captions are the most informative descriptions.
- Multi-label stratification aims to preserve category distributions approximately. It does not guarantee identical distributions between the original dataset and the selected subset.
- Text normalization may remove punctuation that can sometimes contribute to linguistic meaning. Therefore, the original captions are preserved.
- The quality of generated embeddings depends on the selected model and its preprocessing configuration.

These considerations should be taken into account during the implementation and evaluation of the retrieval system.

4.10 Chapter Summary

This chapter presented the main results obtained during Phase 1 and the quality assurance procedures used to examine the prepared dataset.

The results include a 100-image test subset, a standardized caption collection, processed textual descriptions, exploratory data analysis, and multimodal embeddings.

The chapter also discussed the technical challenges encountered during implementation and the limitations that should be considered in subsequent development.

Chapter 5

Conclusion and Future Work

5.1 General Conclusion

This report presented the first phase of the image captioning project, focusing on the exploration, preparation, and transformation of the MS COCO 2017 dataset.

The main objective was to construct a structured and consistent collection of visual and textual information suitable for the development of a Metadata-Filtered Retrieval-Augmented Generation (RAG) system.

The work began with the recovery of the Google Colab environment and verification of persistent storage. The Val2017 dataset was then analyzed to identify different levels of scene complexity and construct a multi-label representation of the 80 object categories.

A stratified sampling procedure was applied to select 100 test images, followed by visual inspection using a contact sheet.

The Train2017 caption annotations were also examined, and a deterministic selection strategy was applied to establish a consistent five-caption structure.

Furthermore, text preprocessing was performed while preserving the original captions. Exploratory data analysis was used to examine caption length and object count distributions.

Finally, image and text embeddings were generated using a CLIP ViT-B/32 model, providing numerical representations that can support similarity-based multimodal retrieval.

Overall, Phase 1 established the data preparation foundation required for the subsequent development of the image captioning system.

5.2 Main Contributions

The principal contributions of this phase can be summarized as follows:

- Organization of the project environment and persistent dataset storage.

- Analysis of scene complexity using object counts and category diversity.
- Construction of a multi-label matrix representing the 80 COCO object categories.
- Selection and visual inspection of a 100-image test subset.
- Standardization of caption annotations using a deterministic five-caption selection policy.
- Text preprocessing with preservation of the original human-written captions.
- Exploratory analysis of textual and visual dataset characteristics.
- Generation and storage of multimodal image and text embeddings.

These contributions provide the structured data resources required by the next stage of the project.

5.3 Future Work

The next phase will focus on using the prepared resources to implement the Metadata-Filtered RAG architecture.

The planned activities include:

1. Implementing the retrieval mechanism using the generated multimodal embeddings.
2. Integrating metadata-based filtering into the retrieval process.
3. Retrieving relevant images and their associated captions according to an input query.
4. Connecting the retrieval component to the generation component.
5. Evaluating the system using the common test subset and its reference captions.
6. Comparing the system's performance using appropriate image captioning and retrieval evaluation metrics.

Particular attention will be given to retrieval quality, caption relevance, and the ability of the system to handle images representing different levels of visual complexity.

5.4 Final Remarks

The preparation of a reliable dataset is a fundamental requirement for building an effective image captioning system.

By organizing the source data, standardizing captions, analyzing image characteristics, and generating multimodal embeddings, this phase provides the foundation for the implementation of the Metadata-Filtered RAG approach.

The prepared resources will serve as the starting point for the next phase, where the retrieval and generation components will be developed and evaluated.